

Attribution-Aware Bayesian Memory Governance

Follow-up Experiment Extension

Revised E1–E4 plan after the five-branch structural-credit audit

Companion extension to the systematic research report dated 21 September 2026

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Source basis: supervisor-approved ABMG step-by-step guide; ABMG systematic research report (21 Sep 2026); Last Paper research handoff; `test_phase3_AIF_simplified_v22.py`; and GitHub branch `stage1-factorizability-audit` at commit `c1c690d80b61694116e0d8b64a4a4b1a2548f0fe`.

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1 Purpose and status of this extension

This document extends the planning sections of the 21 September 2026 ABMG systematic research report. It does **not** replace the original C1–C5 research claims, the frozen-backbone principle, or the already completed Stage-1 evidence. The purpose is narrower: convert the latest discussion into an executable sequence of follow-up experiments, with explicit controls, candidate algorithms, leakage boundaries, metrics, and stop/go rules.

Concept: Architecture preservation

The deployed visual backbone remains frozen online. Learning remains in external state: normal-state belief, factor/source attribution state, correction/spatial experience state, and later the query controller. This extension refines how those states are tested; it does not redefine the overall research paradigm.

The main changes relative to the previous revised roadmap are:

- Remove the former “E0” as an experiment. A procedural snapshot of the completed evidence is sufficient.
- Split spatial follow-up into **E1A localization diagnosis** and **E1B rare-mask utility**.
- Broaden **E2** from a single FactorVAE candidate to a controlled address-representation benchmark.
- Broaden **E3** from NIG alone to a compact normal-state model audit, while retaining NIG as the canonical proposal-aligned Bayesian model.
- Reframe **E4** as **asymmetric feedback utilization**: verified normal and verified defect feedback should produce different, evidence-appropriate persistent writes.

2 Evidence inherited from completed experiments

The follow-up work should start from what has already been established rather than re-running completed causal controls.

2.1 Structural credit is already supported source-side

The five-branch experiment held evidence, support, stream order, query locations, revealed labels, scorer, sentinel set, and checkpoints fixed while changing only write credit. The Sensor condition after 32 feedback events produced approximately:

Condition	Sensor macro-F1
No update	0.3957
Mass-matched broadcast	0.4254
Full broadcast	0.4259
Correct address	0.6631
Shuffled/wrong address	0.1215

The current supported interpretation is therefore:

write destination matters; more write mass is not equivalent to correct structural credit.

The correct-address branch also preserved exact persistent-state locality in the sequential audit, although behavioral interference was not zero because factor scores compete. This is best described as *localized credit with bounded collateral competition*.

2.2 Representation and localization conclusions carried forward

The tested whole-feature beta-VAE reconstruction path damaged anomaly ranking and did not produce the desired low-correlation organization. That is a negative result for one architecture, not proof that address representations are unnecessary.

Raw frozen DINO evidence nonetheless showed reusable cross-product defect-source structure, and the shared-diagonal shrunk-memory scorer substantially outperformed simple centroid and exemplar-max alternatives in the addressability audits. At the same time, Oracle-localized evidence remained materially stronger than Sensor-selected evidence. This Oracle–Sensor gap is the strongest current upstream deployment bottleneck.

2.3 Claims that remain genuinely open

The following should still be treated as hypotheses rather than established conclusions:

- source-enriched C3 feedback improves practical factor credit;
- the full Normal-Inverse-Gamma (NIG) normal-state model is superior under sparse/noisy conditions;
- rare mask requests have persistent value beyond the queried item;
- an address sidecar is required beyond raw frozen DINO;
- active query allocation improves the human-cost/performance frontier;
- the integrated mechanism generalizes across all five sealed target folds and beyond DINOv2 + Real-IAD.

3 Revised immediate experiment map

The immediate sequence is now:



Stage	Main question	Core comparison	Exit criterion
E1A	Where does Sensor localization lose evidence relative to Oracle?	Fixed Sensor vs offline Oracle diagnostics	Failure mechanism quantified
E1B	Can rare human masks produce persistent future localization gain?	No mask vs random-mask vs uncertainty-selected mask at matched budgets	Future patch purity/recall and downstream routing improve
E2	Which downstream representation gives stable factor addresses without replacing the detector?	Raw DINO baseline vs compact sidecar candidates	Better addressability without detector utility loss
E3	Which sparse normal-state model best supports stable surprise, uncertainty, and selective updates?	Deterministic Gaussian, NIG, NIW, hierarchical NIG, optional robust extension	Source-side evidence identifies a justified state model; NIG remains canonical C4 model
E4	How should verified normal and verified defect feedback produce precise asymmetric writes?	Normal-side and defect-side ablations under identical query events	Correct feedback use improves future behavior with bounded collateral effects

Table 1: Immediate follow-up plan. Later E5+ stages remain provisional and are intentionally not redesigned here.

4 E1: spatial evidence and rare-mask learning

4.1 E1A: formal Oracle–Sensor localization-gap audit

Question. Exactly why does deployment-style Sensor evidence underperform Oracle-localized evidence?

E1A is diagnostic only. It must not train a new model. Ground-truth masks are used offline to measure failure, never exposed to the online learner.

For each defective item, retain the current patch ranking and report at least:

$$\begin{aligned}
 \text{Hit@K} &\in \{0, 1\}, \\
 \text{Precision@K} &= \frac{\#\{\text{selected patches overlapping defect}\}}{K}, \\
 \text{Recall@K} &= \frac{\#\{\text{defect patches selected}\}}{\#\{\text{defect patches}\}}.
 \end{aligned}$$

Add:

- first defect-patch rank;
- weighted defect mass under the current patch-responsibility distribution;
- pixel AUROC and AUPRO;
- error taxonomy: ranking error, dilution, patch-resolution error, contextual contamination, and insufficient normal-support coverage;
- downstream routing degradation attributable to each localization failure category.

Decision rule. E1A exits when the Oracle–Sensor gap is decomposed into measurable failure modes rather than reported as a single performance difference. No learning mechanism should be chosen before this diagnosis is complete.

4.2 E1B: rare-mask utility as a costly human action

The revised ABMG report proposes a mask as a costly action, selected only when its expected future value is high. E1B should test this claim directly while keeping the principal experiment compatible with the original guide, in which masks are normally withheld from the online learner.

Define spatial responsibility over candidate patches:

$$q_{t,p} = \frac{\exp(\tau u_{t,p})}{\sum_k \exp(\tau u_{t,k})},$$

and spatial uncertainty:

$$H_{\text{spatial}}(t) = - \sum_p q_{t,p} \log q_{t,p}.$$

Candidate mask budgets should be predeclared, for example:

0%, 0.25%, 0.5%, 1% of stream items.

4.2.1 Matched E1B conditions

ID	Mask policy	Purpose
B0	No masks	Deployment baseline
B1	Random mask requests	Tests the value of the same mask budget without intelligent selection
B2	High- H_{spatial} mask requests	Tests whether localization uncertainty identifies high-value mask opportunities
B3	High- H_{spatial} masks, read-only	Mask may fix/inspect the current item but is not written into persistent state; isolates persistent-learning value

Only when a queried defective item is selected for a mask action should the simulated operator reveal its ground-truth mask. Masks for all other samples remain evaluation-only.

4.2.2 Primary E1B outcome

A mask is useful only if it improves later samples. Therefore the main outcomes are:

- future Sensor patch precision/recall and weighted defect mass;
- downstream factor-routing F1/margins on never-updated sentinel samples;
- reduction in future mask requests required to reach a target localization quality;
- collateral false-positive change on normal samples;
- gain per mask, not merely gain per queried item.

Decision rule. Retain rare-mask learning only if a small budget produces persistent future benefit. If B2 does not beat B1, keep masks as a possible intervention but do not claim that spatial entropy is a useful mask-selection signal.

5 E2: address-representation candidate audit

5.1 Design principle: separate detection geometry from address geometry

The detector should remain in raw frozen DINO space. A candidate address representation is allowed to organize evidence for memory routing, but it must not be required to reconstruct and replace the anomaly detector representation.

$$\text{raw DINO } h \xrightarrow{\text{detector}} s^{AD}(x), \quad h \xrightarrow{\text{optional sidecar}} z^{addr} \rightarrow \text{factor responsibility.}$$

This separation is a direct response to the whole-feature beta-VAE negative result.

5.2 Candidate set

The candidate family should be broader than FactorVAE but still bounded enough to preserve a mechanism-first study.

Candidate	Why test it	What would count as useful evidence?
Raw DINO / no sidecar	Mandatory baseline; current audits already show reusable structure	Addressability is already adequate without additional representation learning
PCA / ICA	Cheap linear sanity check	Stable address structure can be obtained without a deep generative model
Beta-TCVAE	Penalizes total correlation more selectively than multiplying the full VAE KL term	Better addressability than plain beta-VAE without the same information loss
FactorVAE	Explicitly pressures aggregate-posterior factorization	Improved source/aspect stability and separation
CorrVAE-inspired model	Allows structured correlation instead of demanding full coordinate independence	Stable address groups when meaningful properties remain correlated
Address-plus-residual autoencoder	Separates addressable code from residual visual detail	Stable address code while residual channel preserves information
Sparse autoencoder (exploratory)	Tests sparse activation as an address mechanism without a variational bottleneck	Repeatable sparse addresses across products with good locality

5.3 Training and leakage boundary

- Target-category labels, source labels, and masks remain sealed for representation fitting and hyperparameter selection.

- Sidecars are trained only from source-side data under the predeclared fold protocol. The default unsupervised setting should use source normal descriptors unless a method explicitly requires a different source-only objective.
- Defect-source labels may be used for *offline source-CV addressability evaluation*, not to train the target-time representation.
- The raw DINO anomaly score remains unchanged across candidates so addressability is not confounded with detector replacement.

5.4 E2 metrics

A candidate is judged by a multi-objective addressability panel:

- cross-product source/aspect separability;
- within-source responsibility similarity and across-source separation;
- stability across support seeds and product categories;
- spatial consistency with offline masks;
- routing performance under the existing sparse-memory audit;
- dimensional efficiency / active-address count;
- no degradation of the raw detector, because the detector remains raw DINO by construction.

Decision rule. If raw DINO provides competitive stable addresses, select no sidecar. A learned sidecar survives only if it adds addressability that is reproducible across held-out source products and does not require target leakage.

6 E3: sparse normal-state model audit

6.1 Why broaden E3 without abandoning NIG

The supervisor-approved C4 hypothesis specifically compares deterministic Gaussian normality with NIG/Student- t Bayesian normality. NIG therefore retains special status as the canonical proposal-aligned Bayesian model. However, E3 may explore a compact set of alternative normal-state models on source validation to determine whether diagonal independence, shared priors, or robustness assumptions materially affect sparse online governance.

The research question is:

How should sparse verified-normal evidence represent normality and uncertainty?

6.2 Candidate normal-state models

Model	State assumption	Scientific role
Deterministic diagonal Gaussian	Running mean/variance plug-in estimates	Mandatory simple comparator
Diagonal NIG / Student- t	Mean and variance uncertain per coordinate	Canonical proposal-aligned Bayesian model

Normal-Inverse-Wishart (NIW)	Mean and full covariance uncertain within a low-dimensional factor group	Tests whether within-factor correlations matter
Hierarchical NIG	Factor/category posteriors share source-derived hyperpriors	Tests whether structured prior transfer improves ultra-few-shot stability
Robust heavy-tail extension (optional)	Normal evidence may contain occasional contamination/outliers	Robustness test only if E3 diagnostics show sensitivity
Mixture model (deferred)	Multiple normal modes	Introduce only if diagnostics prove persistent multimodality not handled by simpler state/memory modes

For one scalar coordinate, the canonical NIG model remains:

$$z \mid \mu, \sigma^2 \sim \mathcal{N}(\mu, \sigma^2), \quad (\mu, \sigma^2) \sim \text{NIG}(\mu_0, \kappa_0, \alpha_0, \beta_0).$$

Verified-normal structural responsibility supplies an effective fractional write weight:

$$w_{t,g} = r_{t,g}.$$

Each candidate model should expose the same conceptual interface:

`update(z, w), predictive_log_prob(z), uncertainty(), state_drift().`

6.3 Matched E3 stress matrix

All candidates should see the same support sets, streams, query indices, factor credit, and correction rules. Stress conditions should include:

- 1-, 2-, and 4-shot normal initialization;
- random support seeds;
- rare normal modes / context shifts;
- small numbers of queried normals;
- controlled feedback noise;
- long recurrence gaps;
- bounded memory pressure;
- deliberately correlated factor-group coordinates when evaluating NIW.

Primary outputs are calibration, false-positive behavior, posterior/state drift, predictive surprise stability, safe write behavior, and downstream anomaly utility. E3 is not allowed to turn into an unrestricted density-estimator competition.

Decision rule. Use E3 to choose a defensible implementation default for later integration, but preserve the proposal-aligned deterministic-vs-NIG comparison as a confirmatory C4 experiment. Alternatives are secondary unless they expose a specific assumption failure.

7 E4: asymmetric utilization of verified human feedback

7.1 Core principle

Verified normal and verified defect observations do not carry symmetric information.

A verified normal tells the learner that the entire item is legitimate normality; if the detector found it suspicious, the observation is especially informative about false-positive structure. A verified defect does *not* justify expanding normality and, without a mask, does not prove that every selected patch is defective.

Therefore E4 should test an asymmetric write policy:

$$y_t = 0 \Rightarrow \text{normal-state update} + \text{negative/hard-normal correction}$$

$$y_t = 1 \Rightarrow \text{source-to-factor attribution} + \text{positive correction, without expanding normality}$$

7.2 Verified-normal branch

7.2.1 A1. Factor-addressed normal-state update

For a verified normal item, factor responsibility governs how strongly each normal-state component learns:

$$B_{g,t+1}^N = \text{Update}(B_{g,t}^N, z_{t,g}, r_{t,g}).$$

This is the proposal-aligned C2 mechanism: highly responsible factors absorb more of the newly verified normal variation, while unrelated factors move less.

7.2.2 A2. Hard-normal / negative-correction memory

A high-anomaly patch inside a verified-normal image is unusually valuable because its label is unambiguous: it is a false-positive visual pattern under the current detector.

A simple correction form is:

$$s_{t,p}^{new} = s_{t,p}^{(0)} - \lambda_N N_{t,p} + \lambda_C C_{t,p},$$

where $N_{t,p}$ represents similarity to previously verified hard-normal/nuisance evidence and $C_{t,p}$ is a bounded correction term.

The older v22 system already contained the useful memory primitives for this role: high-plasticity correction bank, STM, mature bank, and low-plasticity LTM. The follow-up implementation should reuse the principle but make promotion/retention rules explicit and testable.

7.2.3 A3. Should normal samples have their own recurring pattern memory?

Possibly, but not as one global normal centroid. Normality is likely multimodal. A single

$$q_N = \frac{1}{N} \sum_t r_t$$

can average distinct legitimate modes into a pattern that represents none of them.

A safer exploratory mechanism is a **conditional hard-normal mode memory**, for example:

$$q_m^N = \mathbb{E}[r_t \mid y_t = 0, \text{hard-normal mode } m],$$

or a context-conditioned version q_c^N when product/view/context is known. Such a mechanism should be introduced only if E3/E4A diagnostics show recurring hard-normal modes not already handled by the normal-state model.

7.3 Verified-defect branch

7.3.1 B1. Source-to-factor attribution memory

For queried defects with human source label d , maintain the running responsibility centroid:

$$q_{d,t} = \frac{1}{n_{d,t}} \sum_{i < t: d_i = d} r_i.$$

The source label does not name a latent factor. It regularizes the current noisy responsibility toward the recurrent attribution pattern of that manifestation.

Binary-only credit is:

$$\alpha_t^{bin} = r_t.$$

Source-enriched credit is:

$$\alpha_t^{src} = \eta_s r_t + (1 - \eta_s) q_{d_t, t}.$$

A shuffled-source negative control uses $q_{\pi(d_t), t}$ instead.

7.3.2 B2. Positive correction/write assignment

For a false negative, the factor correction can use the source-enriched credit:

$$\delta_{g,t+1} = \text{clip}(\delta_{g,t} + \eta_\delta \omega_t \alpha_{g,t}, -\delta_{\max}, \delta_{\max}),$$

with $\omega_t = +1$ for false-negative correction and $\omega_t = -1$ for false-positive correction.

The purpose of q_d is not to model the infinite abnormal manifold. It is to reduce variance in *where the correction is written*. The completed five-branch experiment already established why this matters: correct destination helped strongly; deliberately wrong destination caused severe negative transfer.

7.3.3 B3. Interaction with masks

Without a mask, verified-defect feedback is spatially weak: at least one patch is defective, but the learner should not assume that all top- K patches are positive defect evidence. If a rare mask is available, positive correction can be spatially restricted to annotated defect patches and the corresponding factor responsibilities.

7.4 Outcome-aware write strength

The four operating outcomes carry different information:

Outcome	High-value update	Caution
FP	Strong normal-state evidence plus negative/hard-normal correction	Avoid global suppression of unrelated factors
TN	Normal-state evidence; usually weaker correction value	Do not overpopulate correction memory with easy normals
FN	Strong positive correction plus source attribution; mask especially valuable if localization is uncertain	Do not expand normality or treat every top- K patch as defective
TP	Source-attribution consistency and possibly weak positive correction	Detector was already correct; correction value may be smaller than FN

Outcome-dependent strengths should be tested rather than assumed to be optimal.

7.5 E4 decomposition

To keep causal interpretation clean, E4 should itself be staged.

7.5.1 E4A: verified-normal utilization

Keep defect handling fixed. Compare:

- N0: factor-addressed normal-state update only;
- N1: N0 + hard-normal negative correction/STM;
- N2 (exploratory): N1 + conditional recurring hard-normal mode memory, only if diagnostics justify it.

Primary question: does explicit hard-normal memory reduce future false positives without degrading anomaly recall?

7.5.2 E4B: verified-defect source enrichment

Freeze the best justified normal-side mechanism from E4A. Compare:

- D0: binary-only factor credit r_t ;
- D1: correct source-enriched credit α_t^{src} ;
- D2: shuffled-source credit using a fixed source permutation;
- D3: oracle-address upper bound for diagnostic purposes only.

Measure credit quality, same-source future benefit, off-source collateral movement, harmful flips, and final operating metrics.

7.5.3 E4C: combined asymmetric feedback system

Only after E4A and E4B survive separately should the best justified normal and defect branches be combined. This prevents one branch from hiding failure of the other.

Decision rule. C3/source memory is retained only if correct source labels outperform both binary-only and shuffled-source controls at identical query indices. Hard-normal memory is retained only if it improves future false-positive behavior without unacceptable anomaly-recall loss.

8 Implementation plan in the current repository

The current active research branch is `stage1-factorizability-audit`, whose latest commit adds the five-branch confirmation protocol. The follow-up should start from that checkpoint on a new branch rather than modifying historical Stage-1 scripts in place.

A reasonable branch name is:

`stage2-followup-credit-governance`

Proposed bounded implementation units:

File	Responsibility
<code>abmg_localization_gap_audit.py</code>	E1A metrics and failure taxonomy; no learning
<code>abmg_rare_mask_utility.py</code>	E1B matched mask budgets and persistent spatial-memory ablation
<code>abmg_address_sidecar_benchmark.py</code>	E2 memory interface for raw-DINO, linear, and learned sidecars
<code>abmg_normal_state_models.py</code>	Reusable deterministic Gaussian, NIG, NIW, hierarchical-NIG state objects
<code>abmg_normal_state_audit.py</code>	E3 matched source-side stress tests
<code>abmg_feedback_write_audit.py</code>	E4A/E4B/E4C asymmetric feedback conditions
<code>tests/test_...</code>	Focused unit tests for every state update, credit vector, write footprint, mask budget, and leakage boundary

Existing scripts should be reused rather than copied where possible, especially the current patch aggregation, memory addressability, sequential stream/sentinel split, four/five-branch update-integrity checks, and v22 correction/STM primitives.

9 Experimental discipline and non-negotiable controls

1. **Fixed-query replay for mechanism tests.** E1–E4 comparisons that assess memory mechanisms must use identical query events. Endogenous query policies remain a later C5 problem.
2. **Target categories remain sealed.** Representation/model selection uses source-side cross-validation only. Target labels and masks may be used only for final evaluation according to the frozen outer-fold protocol.
3. **Masks are costly feedback, not free training labels.** Offline masks may evaluate every item; online masks are revealed only when the experimental mask policy spends budget.
4. **One declared mechanism per causal comparison.** Do not simultaneously change representation, normal-state model, query sequence, and correction policy in one confirmatory branch.
5. **Negative results are retained.** A component is not rescued by adding complexity after its gate fails.
6. **No semantic/causal claims without evidence.** A factor remains an operational address unless semantic or causal interpretation is independently demonstrated.

7. **No abnormal-manifold assumption.** Source-to-factor memory regularizes attribution/write destination; it is not a required density model of all possible defects.

10 Immediate stop/go matrix

Question	Proceed if	If not
Does the localization audit reveal actionable Sensor failure modes?	Error categories explain a substantial part of Oracle–Sensor degradation	Keep Sensor fixed and avoid speculative spatial complexity
Do rare masks amortize?	Tiny mask budgets improve later localization/routing beyond the queried sample	Remove mask-learning claim; masks may remain diagnostic only
Does entropy-based mask selection help?	High-entropy selection beats random at equal budget	Use simpler/random mask allocation until C5
Does an address sidecar help?	Reproducible addressability gain beyond raw DINO with no target leakage	Keep raw DINO/no sidecar
Does a richer normal-state model help?	Better sparse/noisy calibration, state stability, or write decisions under matched evidence	Retain the simpler model; still run proposal-aligned NIG C4 comparison
Does hard-normal memory add value?	Future FP reduction without unacceptable anomaly-recall loss	Use normal-state update alone
Does correct source memory improve credit?	Correct-source beats binary-only and shuffled-source controls	Narrow/remove C3 source-enrichment claim

11 What is intentionally postponed

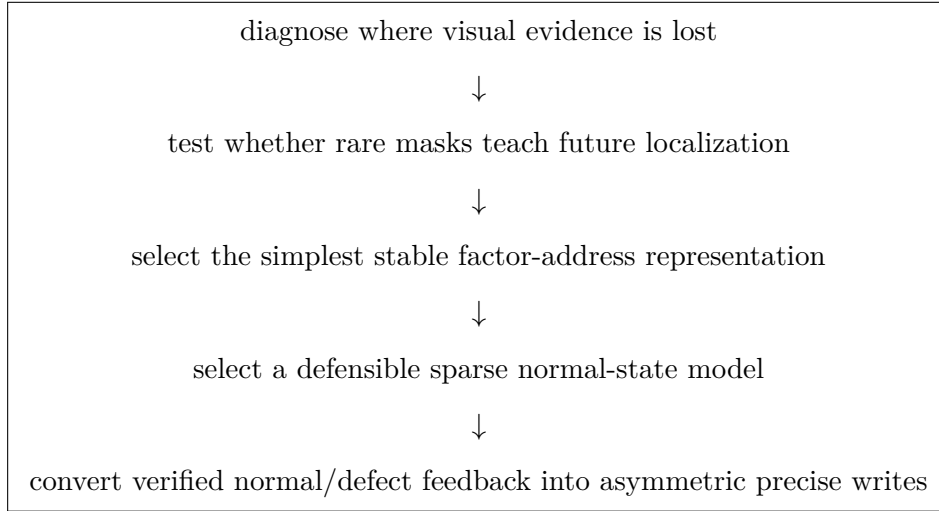
This extension stops at E4. The later ideas in the 21 September report remain valid hypotheses but should not be over-specified before E1–E4 produce evidence. In particular, the following are postponed:

- full joint spatial-structural credit $c_{t,p,g} = q_{t,p}r_{t,g}$;
- the final confirmatory deterministic-vs-NIG C4 experiment after the representation and feedback mechanisms are frozen;
- C5 query-value / active-inference controller;
- full five-fold sealed-target rotation;
- second-backbone / second-dataset robustness matrix.

The next planning revision should occur only after the E1–E4 evidence is available.

12 Compact operational summary

The follow-up research should now be guided by the following chain:



The central scientific invariant remains unchanged:

observation → surprise/responsibility → human evidence → credit assignment → appropriate external-state up

The research should remain mechanism-first and evidence-governed. Correct structural write destination is already supported under the source-side diagnostic. The next objective is to determine how deployable visual evidence and sparse human feedback can estimate that destination, spatially and structurally, with sufficient precision to improve future inspection behavior without contaminating unrelated memory.

A Source mapping for this extension

This extension is grounded in the following project materials:

- *Attribution-Aware Bayesian Memory Governance - Step-by-Step Guide*: supervisor-approved architecture, C1–C5 definitions, factor-addressed normal update, C3 source centroid, bounded factor correction, fixed-query discipline, and NIG C4 comparator.
- *ABMG Systematic Experiment Report, Current Conclusions, and Revised Research Roadmap* (21 Sep 2026): completed Stage-1 results, Oracle–Sensor bottleneck, rare-mask proposal, sidecar candidates, memory hierarchy, and stop/go rules.
- *Last Paper - Research Session Handoff* (15 Sep 2026): experimental history and research-discipline constraints.
- `test_phase3_AIF_simplified_v22.py`: correction-bank, STM/mature/LTM, signed correction, and outcome-aware memory primitives used as design references rather than copied assumptions.
- GitHub repository <https://github.com/jiangdizhao/ATTRIBUTION-AWARE-BAYESIAN-MEMORY-GOVERNANCE> branch `stage1-factorizability-audit`, including the latest five-branch structural-credit protocol and existing audit scripts.

B Notation introduced or reused

Symbol	Meaning
h	Frozen DINO feature / patch descriptor
z^{addr}	Optional address-sidecar representation
$q_{t,p}$	Spatial responsibility of patch p for item t
$H_{\text{spatial}}(t)$	Entropy of spatial responsibility; localization uncertainty
$r_{t,g}$	Structural/factor responsibility assigned to factor address g
B_g^N	Normal-state belief for factor g
$q_{d,t}$	Running source-to-factor responsibility centroid for human source label d
α_t^{bin}	Binary-only structural credit, equal to current r_t
α_t^{src}	Source-enriched structural credit combining r_t and $q_{d,t}$
$\delta_{g,t}$	Bounded additive factor-level correction state
$N_{t,p}$	Similarity/consistency with verified hard-normal or nuisance evidence
η_s	Mixing coefficient between current responsibility and source memory
η_δ	Correction learning rate
ω_t	Signed correction direction; negative for false-positive correction, positive for false-negative correction